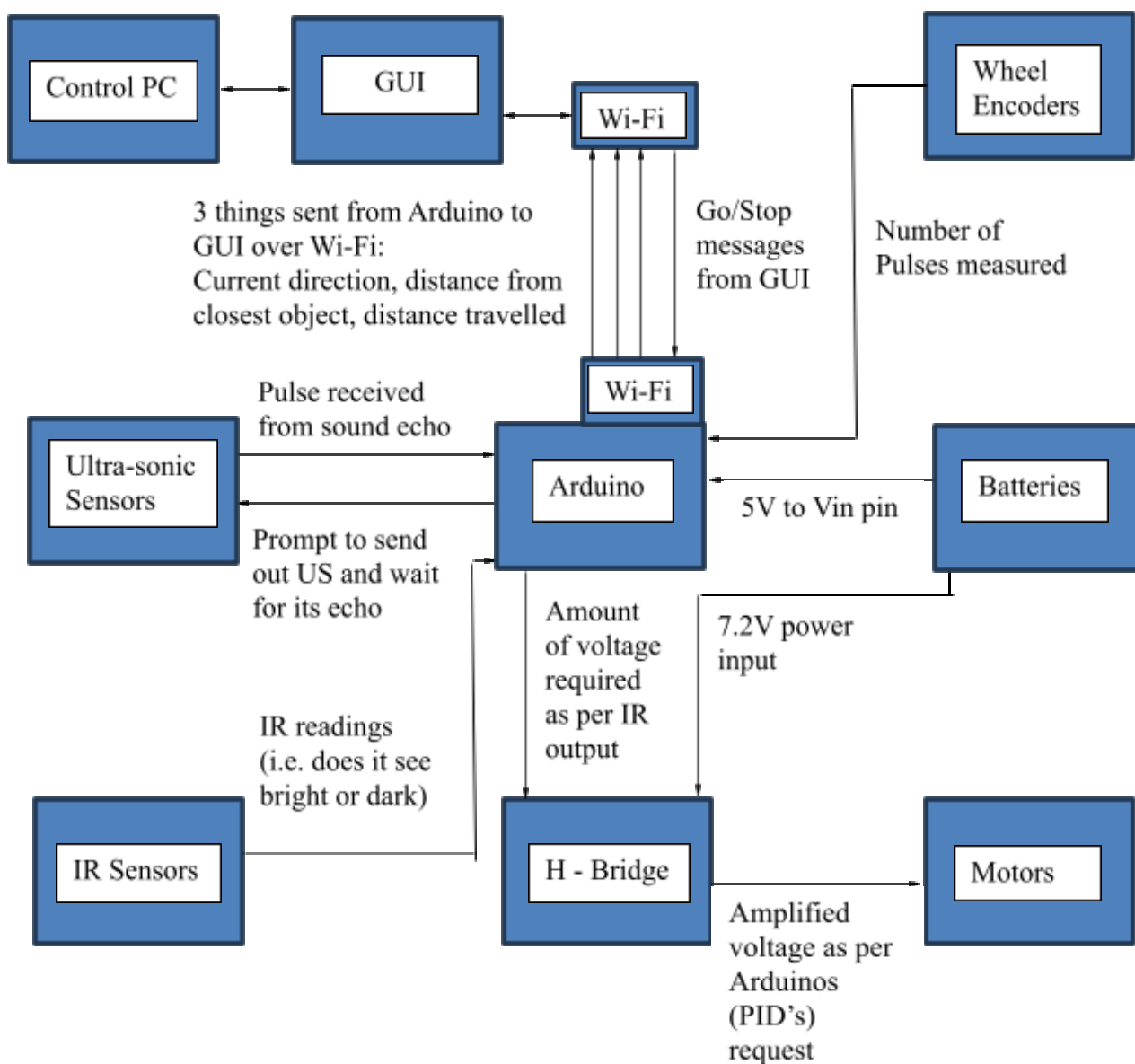




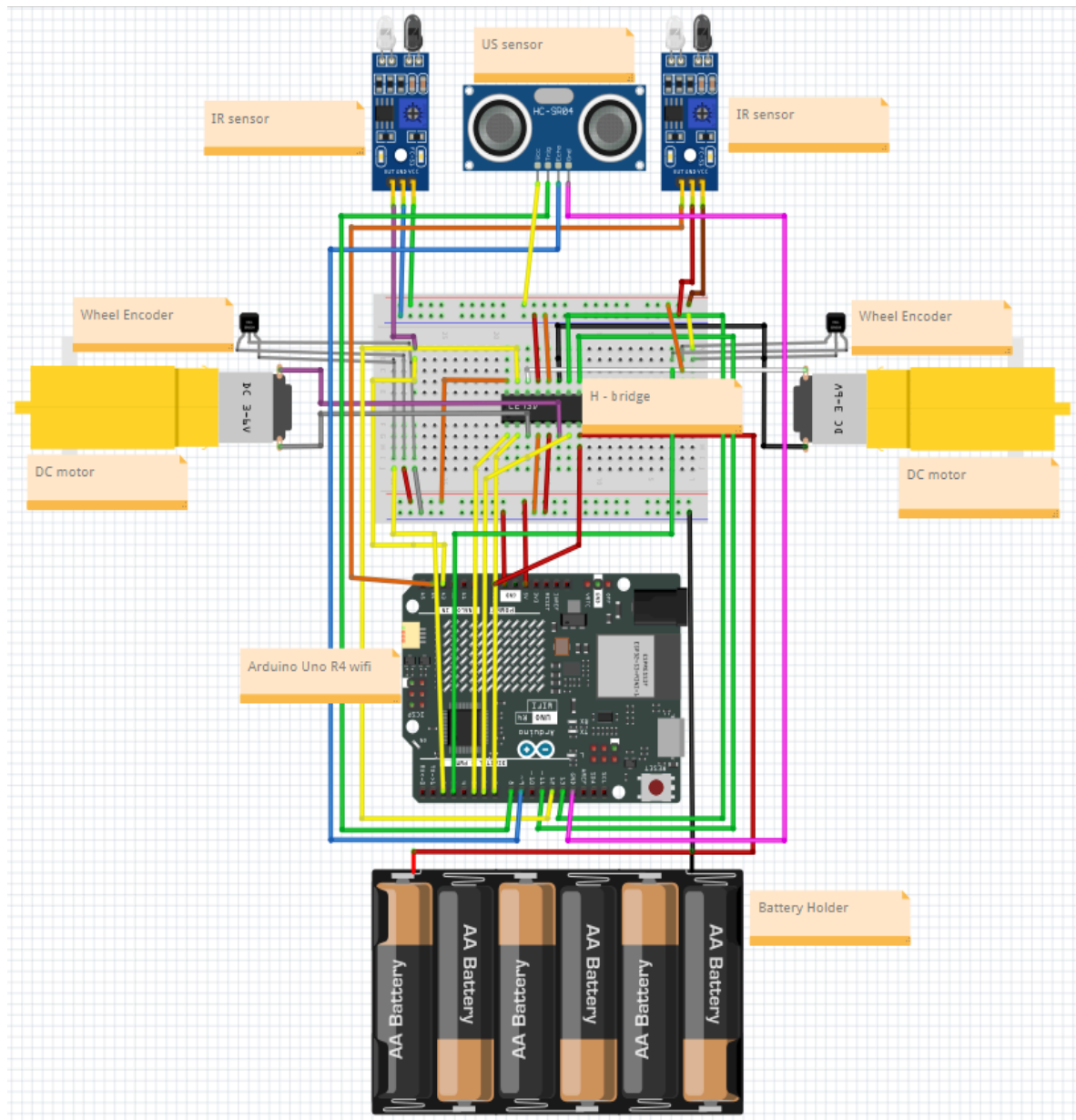
2E10 Final Report (Part B)

Group X15

Block Diagram for Design Flow



Fritzing schematic & diagram



The layout of our breadboard is the exact same as it was for the bronze challenge, as absolutely zero changes had been made. As such, the fritzing schematic of our circuit design is identical to what it was for our interim report. Although we did intend on, we never got around to tidying up the wiring of the buggy, and as a result it is still just as aesthetically displeasing as it has been.

Although the circuit presented is the exact same as it was, I will still explain how it achieves the required functionality.

1 - Follow a white line using the IR sensors:

As seen above, both IR sensors are connected to ground, 5v power, and a pin on the Arduino. This means that the Arduino can respond to the sensor's states and move the motors according to the logic embedded on it. This is true for both the object following mode as well as the reference speed mode, as the buggy will follow the white line as long as it is within the operational range of speeds.

2 - React to and follow objects in front of the buggy using the UltraSonic sensor:

Using the four pins on the US sensor (5v power, ground, trigger and echo), we calculated the distance from the nearest object and acted accordingly. This meant that the buggy stopped completely when within 10cm of the object, and if it was set to object following mode it would follow it at 15cm, with an upper limit of 30cm. We had the sensor poll every $\frac{1}{4}$ of a second, which allowed the Arduino to calculate the time elapsed between the trigger signal (output of ultrasonic sound) and echo signal (input of that ultrasonic sound), and hence used this to calculate the distance to the object immediately in front of the sensor.

3 - Measure the distance travelled by the buggy:

It was found that the wheel encoders pulsed approximately 4 times per full revolution of the wheel, and using interrupts we were able to count these pulses and convert them to the distance travelled using the circumference of our buggies wheel.

All of this is powered by six batteries which equates to a total voltage of 7.2V, although we did spend a lot of time using a total voltage of 9V. This is transferred to the H-bridge, which will use all 7.2V, as well as to the Arduino via the Vin pin.

Overall Challenges

The challenges which our group faced concerning the hardware side of the project were not significant. The use of pins was well explained by lectures, and as such wiring everything up correctly was not particularly difficult or time consuming. Even though we had to replace our arduino in week 2 due to a faulty serial port, and our dc adapter due to accidentally shorting the batteries to ground, the issues surrounding our hardware were relatively insignificant, especially when put in the context of the time spent debugging and testing. As such, we can say that we encountered few significant challenges in turning our design into working hardware.